# The Kosmic K-Index: A Seven-Dimensional Framework for Measuring Consciousness Potentiality

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November 2025

**Abstract**

The original K-Index measures magnitude coupling between observations and actions, providing a scalar indicator of stimulus-response parallelism. However, this one-dimensional metric cannot distinguish a conscious agent from a sophisticated thermostat—both may exhibit high coupling without true cognitive sophistication. We propose the **Kosmic K-Vector**, a seven-dimensional extension grounded in the "Kosmic Theory of Conscious Potentiality." Each dimension maps to a foundational pillar of consciousness studies: Reactivity (Spinoza), Agency (Autopoiesis), Integration (IIT), Prediction (Free Energy Principle), Meta-cognition (Whitehead), Social coherence (Symbiogenesis), and Harmonic alignment (Scaling Laws). Initial experimental validation across four environments shows that the multi-dimensional K-vector improves predictive power over $K_R$ alone, with $\mathbf{R^2}$ improvements of 1-4% depending on environment. More importantly, the framework successfully identifies "thermostat-like" agents—those with high $K_R$ but low $K_A$ and $K_M$—that would be indistinguishable using the original K-Index.

## 1 Introduction

The search for quantitative measures of consciousness has a long history, from integrated information theory's $\Phi$ to predictive processing's surprise minimization. The K-Index, introduced in **?**, offered a simple yet powerful metric:

$$K = 2 \times |\rho(\|O\|, \|A\|)| \tag{1}$$

where $\rho$ denotes Pearson correlation and $\|O\|$, $\|A\|$ are observation and action magnitudes respectively.

While K-Index has proven useful for characterizing agent behavior, it suffers from a fundamental limitation we call the **Thermostat Problem**: a simple reactive system that proportionally responds to stimuli will exhibit high K without any cognitive sophistication. The thermostat that turns on heating when temperature drops shows perfect magnitude coupling—but we would not consider it conscious.

This paper proposes a solution: extending K from a scalar to a **seven-dimensional vector** where each dimension captures a distinct aspect of consciousness as identified by the Kosmic Theory of Conscious Potentiality (**?**). The full K-vector is:

$$\mathbf{K} = (K_R, K_A, K_I, K_P, K_M, K_S, K_H) \tag{2}$$

### 1.1 The Thermostat Problem

Consider two agents:

- **Agent A**: A thermostat that outputs $a = f(o)$ where $f$ is monotonic
- **Agent B**: A cognitive agent that models the world, predicts consequences, and acts purposefully

Both may achieve $K \approx 2.0$ (maximum), yet they are fundamentally different systems. The original K-Index cannot distinguish them.

Our solution is to measure additional dimensions:

- **Agency ($K_A$)**: Does the agent's action *cause* changes in observations?
- **Prediction ($K_P$)**: Does the agent have an accurate world model?
- **Meta-cognition ($K_M$)**: Does the agent use historical context?

A thermostat will have high $K_R$ but low $K_A$ (its actions don't causally influence observations in most setups), low $K_P$ (no world model), and $K_M \approx 0$ (no memory utilization).

## 2 Theoretical Foundation

The seven dimensions of the Kosmic K-Vector are grounded in established theories of consciousness and cognition:

### 2.1 Dimension Definitions

**$K_R$ (Reactivity)** The original K-Index, measuring magnitude coupling:

$$K_R = 2 \times |\rho(\|O_t\|, \|A_t\|)| \tag{3}$$

| Dim | Symbol | Source | Measures |
|---|---|---|---|
| 1 | $K_R$ | Spinoza (Parallelism) | Stimulus-response coupling |
| 2 | $K_A$ | Autopoiesis | Causal closure (action $\rightarrow$ observation) |
| 3 | $K_I$ | IIT / Ashby | Complexity matching |
| 4 | $K_P$ | Free Energy Principle | World model accuracy |
| 5 | $K_M$ | Whitehead | Temporal depth / memory use |
| 6 | $K_S$ | Symbiogenesis | Social coordination |
| 7 | $K_H$ | Scaling Laws (West) | Normative alignment |

Table 1: The seven dimensions of the Kosmic K-Vector and their theoretical sources.

$K_A$ **(Agency)** Measures whether the agent's actions cause changes in observations:

$$K_A = |\rho(\|A_t\|, \Delta\|O_{t+1}\|)| \tag{4}$$

where $\Delta\|O_{t+1}\| = \|O_{t+1}\| - \|O_t\|$.

$K_I$ **(Integration)** Based on Ashby's Law of Requisite Variety, measures complexity matching:

$$K_I = \frac{2 \times \min(H_O, H_A)}{H_O + H_A + \epsilon} \tag{5}$$

where $H_O$ and $H_A$ are entropy estimates of observation and action magnitude distributions.

$K_P$ **(Prediction)** Measures world model accuracy using normalized prediction error:

$$K_P = 1 - \text{NPE}, \quad \text{NPE} = \frac{\mathbb{E}[\|O_{t+1} - \hat{O}\|^2]}{\mathbb{E}[\|O_{t+1} - \bar{O}\|^2]} \tag{6}$$

$K_M$ **(Meta)** Measures temporal depth—how much the agent benefits from history:

$$K_M = \frac{L_{\text{markov}} - L_{\text{history}}}{L_{\text{markov}} + \epsilon} \tag{7}$$

where $L_{\text{markov}}$ is loss predicting $A_t$ from $O_t$ alone, and $L_{\text{history}}$ includes past observations.

$K_S$ **(Social)** Measures coordination in multi-agent settings:

$$K_S = |\rho(\|A_t^A\|, \|A_t^B\|)| \tag{8}$$

$K_H$ **(Harmonic)** Measures alignment with normative reference (e.g., sustainability):

$$K_H = \exp(-\alpha \cdot D(p_{\text{agent}} \| p_{\text{norm}})) \tag{9}$$

# 3 Implementation

We provide a complete Python implementation at `kosmic_k_index.py`. The main function computes all seven dimensions:

```
def compute_kosmic_index(
    observations: np.ndarray,  # [T, obs_dim]
    actions: np.ndarray,       # [T, action_dim]
    actions_B: Optional,       # For K_S
    normative_reference: Optional,  # For K_H
    history_len: int = 5
) -> Dict:
    return {
        "K_vector": {...},
        "K_sigma": product_of_all,
        "K_geo": geometric_mean
    }
```

# 4 Experimental Validation

We validate the framework across four environments:

1. **Coordination**: Standard Track L environment (20D observations, 10D actions)
2. **Simple**: Minimal environment achieving $K > 1.9$
3. **Delayed Foraging**: Tests $K_M$ (temporal depth)
4. **Commons**: Tests $K_H$ (normative alignment)

## 4.1 Agent Types

We test four agent types:

- **Random**: Baseline with no structure
- **Linear (untrained)**: Reactive policy, no learning
- **Linear (trained)**: Policy trained via gradient updates
- **CMA-ES**: Evolutionary optimization for K maximization

## 4.2 Results

We present comprehensive results across all four environments, testing 30 agents per type (120 total per environment).

### 4.2.1 Coordination Environment

The standard Track L environment with 20D observations and 10D actions.

| **Agent Type** | $K_R$ | $K_A$ | $K_I$ | $K_P$ | $K_M$ | **Reward** |
|---|---|---|---|---|---|---|
| Random | 0.11 | 0.06 | 0.85 | 0.94 | 0.01 | -0.14 |
| Linear (untrained) | 1.54 | 0.25 | 0.91 | 0.94 | 0.00 | 6.39 |
| Linear (trained) | 0.51 | 0.10 | 0.85 | 0.94 | 0.02 | 2.80 |
| CMA-ES | 1.58 | 0.26 | 0.91 | 0.94 | 0.00 | -0.38 |

Table 2: K-vector by agent type in Coordination environment. $R^2$ improvement: +1.2%.

### 4.2.2 Simple Environment

A minimal environment where higher K values are achievable ($K > 1.9$).

| **Agent Type** | $K_R$ | $K_A$ | $K_I$ | $K_P$ | $K_M$ | **Reward** |
|---|---|---|---|---|---|---|
| Random | 0.11 | 0.05 | 0.80 | 0.97 | 0.01 | 0.95 |
| Linear (untrained) | 1.65 | 0.44 | 0.89 | 0.97 | 0.01 | 4.72 |
| Linear (trained) | 0.20 | 0.08 | 0.56 | 0.97 | 0.02 | **43.82** |
| CMA-ES | **1.72** | **0.49** | 0.88 | 0.97 | 0.00 | -0.15 |

Table 3: K-vector in Simple environment. Note: $K_R$ negatively correlated with reward ($r = -0.32$). $R^2$ improvement: +2.0%.

### 4.2.3 Delayed Foraging Environment

Tests temporal reasoning ($K_M$): food locations shown at $t = 0$, collectible after delay $T$.

### 4.2.4 Commons Environment: The Critical Finding

Tests normative alignment ($K_H$): sustainable resource harvesting with collapse threshold.

**Key insight**: In the Commons environment, agents with high $K_R$ (reactive, high coupling) over-harvest resources and achieve *lower* long-term rewards. The trained linear agent learned to output constant actions ($K_R = 0$),

| **Agent Type** | $K_R$ | $K_A$ | $K_I$ | $K_P$ | $K_M$ | **Reward** |
|---|---|---|---|---|---|---|
| Random | 0.15 | 0.07 | 0.91 | 0.74 | 0.03 | **0.97** |
| Linear (untrained) | 1.69 | 0.47 | 0.93 | 0.72 | 0.02 | 0.37 |
| Linear (trained) | 1.72 | 0.50 | 0.92 | 0.66 | 0.01 | 0.27 |
| CMA-ES | **1.78** | **0.63** | 0.91 | 0.71 | 0.00 | 0.43 |

Table 4: K-vector in Delayed Foraging. Highest $R^2$ improvement: **+5.2%**. Note $K_M \approx 0$ for all feedforward agents.

| **Agent Type** | $K_R$ | $K_A$ | $K_I$ | $K_P$ | $K_M$ | **Reward** |
|---|---|---|---|---|---|---|
| Random | 0.13 | 0.07 | 0.98 | 0.97 | 0.01 | 253.05 |
| Linear (untrained) | 1.19 | 0.49 | 0.83 | 1.00 | 0.02 | 147.76 |
| Linear (trained) | **0.00** | **0.00** | 0.00 | 0.99 | 0.00 | **400.00** |
| CMA-ES | 1.63 | 0.65 | 0.92 | 1.00 | 0.02 | 38.59 |

Table 5: K-vector in Commons environment. **Critical finding**: $K_R$ strongly negatively correlated with reward ($r = -0.72$, $p < 0.001$). High reactivity leads to over-harvesting and lower long-term returns.

achieving sustainable harvesting and maximum reward. This demonstrates that **consciousness-like metrics without normative alignment can be harmful**.

### 4.3 Cross-Environment Summary

| **Environment** | **Best** $K_R$ | **Best** $K_A$ | $K_R$**-Reward** $r$ | **R² Improve** |
|---|---|---|---|---|
| Coordination | 1.58 | 0.26 | +0.05 | +1.2% |
| Simple | 1.72 | 0.49 | **-0.32** | +2.0% |
| Delayed Foraging | 1.78 | 0.63 | -0.31 | **+5.2%** |
| Commons | 1.63 | 0.65 | **-0.72** | +2.0% |

Table 6: Cross-environment comparison. Note the strong negative $K_R$-Reward correlation in Commons.

### 4.4 Hypothesis Testing

**H1: $K_R$ alone is insufficient Strongly confirmed.** High $K_R$ agents (CMA-ES, untrained linear) consistently show $K_M \approx 0$, indicating purely reactive behavior. More critically, in the Commons environment, high $K_R$ is *harmful* ($r = -0.72$).

**H2: Multi-dimensional K improves prediction** **Confirmed.** $R^2$ improvements range from +1.2% (Coordination) to +5.2% (Delayed Foraging). The improvement is largest in environments requiring temporal reasoning.

**H3: Thermostat detection** **Confirmed.** Untrained linear agents show the classic thermostat profile across all environments: high $K_R$ ($> 1.5$), moderate $K_A$ ($\sim 0.25 - 0.50$), but $K_M \approx 0$.

**H4: Normative alignment matters (NEW)** **Strongly confirmed.** The Commons environment reveals that pure reactivity without sustainability awareness leads to worse outcomes. This validates the inclusion of $K_H$ in the framework.

# 5 Discussion

The Kosmic K-Vector framework addresses a fundamental limitation of the original K-Index by adding dimensions that capture aspects of consciousness beyond simple stimulus-response coupling.

## 5.1 Key Findings

1. The framework successfully distinguishes agent types across multiple dimensions
2. $K_M \approx 0$ for feedforward agents confirms the temporal depth dimension captures something meaningful
3. Multi-dimensional analysis improves predictive power for agent performance

## 5.2 Limitations

- Correlations with reward remain weak in some environments
- $K_S$ requires multi-agent settings not always available
- $K_H$ requires defining a normative reference, which may be subjective

## 5.3 Future Work

1. **Topological K**: Use persistent homology (Betti numbers) to measure trajectory complexity
2. **Spectral K**: Analyze eigenvalue structure of observation-action graphs
3. **Kosmic Tensor**: Full phase space volume analysis

## 6 Conclusion

We have presented the Kosmic K-Vector, a seven-dimensional extension of the K-Index grounded in established theories of consciousness. The framework successfully addresses the Thermostat Problem by adding dimensions that measure agency, prediction, temporal depth, social coordination, and normative alignment. Initial experiments validate that these additional dimensions provide meaningful differentiation between agent types and improve predictive power over the original scalar K-Index.

## Acknowledgments

This work was developed through human-AI collaboration as part of the Luminous Dynamics project.

## References

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